Don't Just Tell Me, Ask Me: AI Systems that Intelligently Frame Explanations as Questions Improve Human Logical Discernment Accuracy over Causal AI explanations

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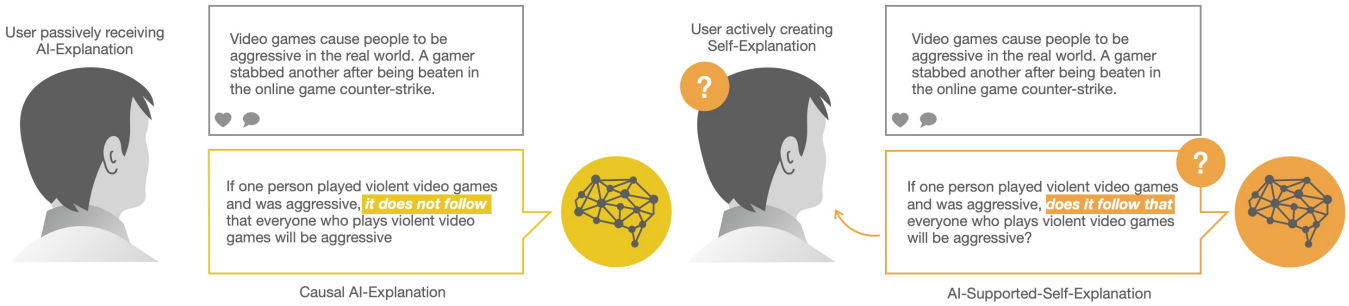


Figure 1: AI systems that ask a user questions can improve human discernment outcomes over AI systems that simply tell people what and why. Left: An example of a socially divisive statement and AI feedback with causal AI-explanations *telling* users why the statement is logically invalid. Right: An example of a socially divisive statement and AI feedback with AI-framed Questioning *asking* the user a question that helps them asses if the statement is logically invalid or not.

ABSTRACT

Critical thinking is an essential human skill. Despite the importance of critical thinking, research reveals that our reasoning ability suffers from personal biases and cognitive resource limitations, leading to potentially dangerous outcomes. This paper presents the novel idea of AI-framed Questioning that turns information relevant to the AI classification into questions to actively engage users' thinking and scaffold their reasoning process. We conducted a study with 204 participants comparing the effects of AI-framed Questioning on a critical thinking task; discernment of logical validity of socially divisive statements. Our results show that compared to no feedback and even causal AI explanations of an always correct system, AI-framed Questioning significantly increase human discernment of logically flawed statements. Our experiment exemplifies a future style of Human-AI co-reasoning system, where the AI becomes a critical thinking stimulator rather than an information teller.

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CCS CONCEPTS

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; **Empirical studies in interaction design**; **Empirical studies in HCI**; **HCI theory, concepts and models**.

KEYWORDS

AI, Human-AI Interaction, Language Model, Explainable AI, AI Explanation, Reasoning, Logic

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1 INTRODUCTION

Artificial intelligence (AI) systems have become ubiquitous in our daily lives impacting every layer of society, from individual decisions on key health issues (e.g. vaccinations), to nationwide security (e.g. voting) [59]. However, these AI models can be biased, deceptive, or appear more reliable than they are, leading to dangerous

decision-making outcomes [29]. For example, an AI system might misinterpret a medical diagnosis due to bias in the data used to train it leading to unnecessary and harmful treatments, a predictive policing system might use faulty data to erroneously target a member of a minority group which can lead to over-policing of marginalized communities, or a social media information recommendation system might falsely claim incorrect or misleading news stories to be true which can lead to political instability and social unrest [17, 18, 46].

This is especially concerning when AI systems are used in conjunction with humans, as people have a tendency to blindly follow the AI decisions and stop using their own cognitive resources to think critically.Prior research shows that people when assisted by AI systems that give them the answers become compliant and stop using cognitive resources to engage in thinking critically about AI-mediated information themselves and take the feedback of the AI at face value [7, 11, 17].

Critical thinking is the ability to logically assess claims, reasons, and beliefs and hold back from making a decision when information is not sufficient or misleading [27]. In the literature, critical thinking has been positively correlated with an increased ability to discern false information from true information [35, 45, 48] and override otherwise intuitive thoughts[13, 25, 34, 35]. As we interact more and more with AI systems in everyday life, we are increasingly exposed to massive amounts of AI-mediated information that can be potentially deceptive, misleading, or strictly false. In such cases critical thinking becomes an important skill to reliably process the information we encounter and integrate it with our existing beliefs and behaviors. However, despite the importance of critical thinking, it is a skill not everyone masters or have the cognitive resources to engage in. Researchers have shown that people frequently use "cognitive shortcuts" that do not involve critical, logical and cautious thinking [15, 24, 30, 47, 51, 56].

While previous work has shown that local explanations with the *causal explanations* strategy (i.e. "X classification because of Y reason") can help people determine the veracity of information, change people's beliefs and improve their decision making *outcomes* [11, 28], using causal AI-explanations does not necessarily improve the human reasoning *process*, as users might just rely on the answers of the AI systems without thinking about the problem for themselves [7, 11, 17]. Complete over-reliance on AI-systems is problematic as (1) it makes users vulnerable to mistakes made by the AI system, and (2) users do not learn how to internalize the skill. Going beyond this challenge and building AI systems that engage the user more deeply to reason for themselves requires development of new human-AI interaction and explanation methods.

This paper presents the novel idea of AI-framed Questioning inspired by the ancient method of Socratic questioning that uses intelligently formed questions to provoke human reasoning, allowing the user to correctly discern the logical validity of the information for themselves. In contrast to causal AI-explanations that are declarative and have users passively receiving feedback from AI systems, our AI-framed Questioning method provides users with a more neutral scaffolding that leads users to actively think critically about information. We report on an experiment with 210 participants comparing causal AI-explanations, AI-framed Questioning, and control conditions' influence on users' ability to discern logically invalid information from logically valid information. Our results show that AI-framed Questioning increase the discernment accuracy for flawed statements significantly over both control and causal AI-explanations of an always correct AI system. We align these results with qualitative reports by the participantsto demonstrate the differences in users thinking processes caused by the questioning method, and discuss generalizability. Our results exemplify a future type of Human-AI co-reasoning method, where the AI systems become critical stimulators rather than a information tellers - encouraging users to make use of their own reasoning and agency potential.

2 RELATED WORK

2.1 Human reasoning and critical thinking

According to existing cognitive models, human reasoning is modulated by the interaction between reflective and intuitive thinking [20, 25, 34, 35]. Intuition often operates as the default mode because it is quick, effortless and automatic without demanding explicit conscious awareness [24]. However, it is prone to errors caused by biases [25, 50] and can easily lead to decision making with unintended or unwanted consequences. With much information consumption and social interaction increasingly happening "online" in closed spheres where individuals are only exposed to like-minded views, intuition-based judgments and decisions will consequently remain unchallenged, biased and even reinforced by their strongly held false beliefs, potentially leading to hatred and hurt [60].

In contrast, reflective thinking is intentional, effortful, and controllable where people consciously make sense of, adapt or justify what they know based on existing and new information [3, 24]. The social aspect of reasoning, such as being challenged by a friend also compels individuals to engage in reflective thinking [20, 32]. When *challenged* by convincing reasoning or questions of someone else, individuals might be provoked to justify their own beliefs. This forces them reflectively find good reasons that can convince others, which, in turn, reduces biases and leads to stronger argumentation, deeper reflection and more optimal information processing and decision making.

In the broader psychology literature, this robust form of reasoning is known as critical thinking, a capability to evaluate the quality of new information (e.g. its logical validity and soundness) and effectively integrate it with one's own beliefs and decision making [16, 53, 54]. The goal of critical thinking is to disrupt/counteract the automatic tendencies towards relying on intuitions (usually driven by old beliefs/biases), and instead, to establish an additional level of thinking, "a powerful inner voice of reason in pursuit of the meaning and truth"[35, 45].

A popular method to help people engage in critical thinking is the Socratic questioning method where instead of one person holding all the knowledge and truth and everyone else listening, the person with the knowledge puts themselves in an ignorant role and collaboratively arrive at the appropriate knowledge through dialogue and framed questioning [44, 45]. In this case the knowledge is arrived at through the people's agency and capacity to identify

contradictions, correct incomplete or inaccurate ideas and eventually discover the fullest possible knowledge, rather than passively relying on the person with knowledge.

Inspired by this method, AI systems could promote users' internal critical thinking by asking them framed questions that assist them at arriving at appropriate knowledge instead of requiring the user to rely on the AI system.

2.2 Technologies for supporting critical thinking

Conversational systems like Alexa, Siri and Google Assistant are playing an increasingly important role in everyday learning and information processing [41]. But while these conversational AI systems are good at extracting relevant information, they have no methods for helping the user evaluate or ask about the information they extract. For example, you cannot ask Alexa about the quality of the extracted information, or why or how it found the information that it did, when the information extracted is potentially vague or misleading. Instead, such conversational systems only provide us with information that can be true or false but they do not necessarily help us to criticize them or boost our ability to think critically. In order to support the general population to be more critical and resilient to information of various quality, recent work has primarily focused on the development of fact-checking algorithms or fake-news detectors [1, 42, 52, 63, 64].

Using fact-checking systems, the truth information can be verified for the users e.g. by checking relevant sources, and they can be provided with additional facts, and opposite opinions in their judgment and decision making. However, research has shown that people often do not bother to check the accuracy of information before sharing it online [57]. Furthermore, even when people are presented with corrective information, they often do not change their beliefs [4, 58]. This suggests that fact-checking in itself is not an effective solution to the problem of misinformation. An alternative approach is to improve human reasoning so that people are better able to discern between true and false information. This has several advantages over fact-checking. First, it has been shown to be more effective in changing people's beliefs than if they are merely presented with corrective information[47]. Second, it does not necessarily rely on people actually checking the factual accuracy of information before sharing it (there is no need for fact-checking the premise of a logically invalid argument). Finally, improved human reasoning should also lead to better decision making in general, which has a range of benefits beyond reducing susceptibility to misinformation.

To this effect, researchers have explored ways in which technologies can teach humans to think more critically such as the construction of chatbots that teach reasoning skills such as fallacy identification[19, 31], probability and uncertainty [61].These approaches have had limited success for three reasons: (1) they require a lot of cognitive effort from the users, (2) the learning examples are often so abstract that there is no guarantee that users will actually be able to use the skills in real-world situations, and (3) the approaches do not work in-situ as the user comes across information in real-life but rather relies on them having learnt them in advance.

Instead, researchers have tried to integrate critical thinking-inducing elements directly into AI systems. For example, Ma and Gajos [37] developed a reversed sequence interface that changes the order of information being presented to the user and found that it significantly reduced biased decision making compared to traditional swipe interfaces. In another work, Pennycook et al. [46] developed a method that simply subtly shifts attention to improve accuracy. Lastly, Danry et al., built a platform-agnostic AI systems that improved people's ability to identify and engage in critical thinking [11].

2.3 Explainable AI: Opportunities and Challenges

With the pervasive adoption of opaque machine learning models in supporting judgment and decision making, the explainability of AI systems has become a critical topic. For everyday lay-users, who may not have deep technical knowledge to understand and use AI in their contexts and avoid AI's mistakes, it is especially critical that AI systems are able to explain their processes effectively [43]. Since the reasoning of a system is often abstract and can be hard to understand, additional research into what makes a proper explanation is required so as to design and engineer AI generated explanations that are more natural and user friendly[9, 14, 33, 43]. If not properly designed and well suited in the context of interaction, AI-generated explanations can be ignored, resisted, or over-relied upon by users. People can develop over-simplified heuristics regarding the AI's competence instead of making efforts to analytically consider each explanation and evaluate its validity and whether it supports the AI's suggestion [2].

To address the problem of over-reliance, researchers have developed explainability methods that cognitively engage the user to think about the AI classification [7, 36]. For instance, Buçinca et al. [7] developed and compared three cognitive forcing functions where the user had limited access to the the AI recommendation and hence would have to rely on their own inferences from information to make a decision. They found that such cognitive forcing functions compelled more thoughtful consideration of AI generated explanations and significantly reduced over-reliance on the AI system in making healthy decisions about food choice. However, the users also experienced these functions as being more cognitively demanding - hindering their desire to use AI systems with such cognitive forcing functions in real-life scenarios.

We believe an AI system that guides the user with intelligently formed questions, could be engage user critical thinking without imposing too strong requirements of cognitive resources. In our work, we seek to evaluate such an approach by investigating the effects of AI-framed Questioning explanations inspired by Socratic questioning on human information discernment.

3 RESEARCH QUESTIONS & DEFINITIONS

Our study aims to explore the effects of AI-framed Questioning on discernment by provoking users with intelligently formed questions when evaluating logical statements around socially divisive topics. These questions serve as a thinking scaffolding for users to evaluate and make decisions for themselves rather than taking the explanations from the AI systems at face value.

In particular, our research questions are:

(1) Do humans perform better at discerning the logical validity of socially divisive statements when they receive feedback from AI systems compared to when they work alone?

(2) How do AI-framed Questioning and causal AI-explanations affect participants' discernment of logical validity, confidence of their discernment, perceived information sufficiency by controlling personal factors (i.e. prior belief, trust in AI, cognitive reflection) as covariates?
(3) Do personal factors, such as prior belief, prior knowledge, trust in AI, cognitive reflection (indicating the level of critical thinking)impact discernment?

From these questions we derive the following hypotheses: (H1) AI and humans together work better than humans alone. (H2) AI framed questioning is more effective than causal explainability and control, and (H3) Personal factors (prior belief, prior knowldge, trust in AI, and cognitive reflection) affects logical discernment accuracy.

In literature, an AI explanation is defined as a description of how an AI system arrives at answers which may vary in *domain*, *strategy*, *content* and *form* [43]. The explanation domain can be either global or local: *Local* AI explanations are defined as "meaningful information about the calculations or logical processes involved in the processing of a particular case" [43] — i.e. information on why the system arrived at *a particular classification*—, while global explanations describe how the AI system arrives at classifications *more generally* (e.g. through a decision tree architecture). The explanation strategy, on the other hand, is about the method with which the content of the domain is being delivered to the user. Examples AI explanation forms include causal explanations [11, 21], analogy-based explanations [23], counter-factual explanations [26], dialogue-based explanations [38, 62], or self-explanations (although never tested with AI-systems) [8]. The explanation content is then the thing that explains a model or specific classification, which can take explanation form as text, statistical graphs, decision trees, feature histograms, color gradients, feature matrices, rule sets.

In literature trust in AI systems is often investigated under various definitions. This paper deploys the definition of trust used in [40] as "the willingness of a [user] to be vulnerable to the actions of [an AI system] based on the expectation that the [AI system] will perform a particular action important to the [user], irrespective of the ability to monitor or control [the AI system]". The definition is assumed in the work of Epstein et al [12] to denote perceived trustworthiness of AI systems.

4 EXPERIMENT

To evaluate the effects of AI-framed Questioning on human discernment, we conducted a 3-by-2 factorial experimental design that asked participants to evaluate the logical validity of socially divisive statements. Participants were randomly assigned into three intervention conditions (between-subjects) including: (1) control condition - no explanation is presented with the statement, (2) "causal AI-explanation" condition - AI provides a intelligently generated causal explanation related to the corresponding statement, (3) "AI-framed Questioning" - AI provides an intelligently adapted question prompting participants to self explain their thinking related to the corresponding statement. Each participant was presented with a series of statements that can be "invalid" or "valid" (within-subject). To control for individual differences, personal factors are measured and analyzed as covariates including prior belief and knowledge on statement topics, trust in AI, and cognitive reflection (see details in 4.5). The study was pre-registered on https://aspredicted.org/L6D_33B under #94860 before being conducted.

4.1 Materials

The statements used as stimuli in this study came from the "IBM Debater - Claims and Evidence" dataset [49], which contains both labeled claims and labeled evidence for 58 different socially divisive topics, such as 'immigration', 'poverty', 'secular societies', etc. The claims and evidence have been labelled thematically in advance by the authors of the original dataset to make up a total of 4,692 statements of claim+evidence pairs with evidence types being 'study', 'expert' and 'anecdotal' evidence.

Given this dataset, we sampled five topics randomly: (1) "violent video games cause aggression", (2) "affirmative action counters the effects of a history of discrimination", (3) "refugees should be embraced", (4) "Israel should lift the blockade of Gaza", and (5) "male infant circumcision should be less prevalent". Within these topics we sampled five anecdotal claim+evidence pairs and five non-anecdotal claim+evidence pairs randomly from the dataset for each topic (50 in total). As known in literature, statements that uses *anecdotes* to support their claims suffer from the hasty generalization fallacy by making a general claim based on only one particular instance ("One X therefore all X") [22]. The anecdotal claims+evidence pairs were thus labeled "logically invalid" and the non-anecdotal pairs were labeled "logically valid". Next, we verified the logical validity of each of the statements. A statement is defined as logically valid if and only if it is impossible for the reasons in a statement to be true and the conclusion false. Hence, the main claim of a statement can be false while the statement can be logically valid. It is not required for a valid statement to have reasons that are actually true, but to have reasons that, if they were true, would guarantee the truth of the statement's conclusion. Conversely, a statement is logically invalid if and only if the reasons in a statement can be true and while the conclusion is false. For example, the statement "I have an orange box. I know all orange boxes contain pears. Therefore my orange box contains pears." is a valid statement. If it true that I have an orange box and that all orange boxes contain pears then the conclusion that my box contains pears must necessarily also be true. Conversely, if I don't have an orange box, or all orange boxes do not contain pears, my orange box would also not necessarily contain pears. In contrast, the statement"I have an orange box, and it doesn't contain pears. Therefore goats orbit Saturn." is an invalid statement (there is no logical link between my orange box without pears and whether or not goats are orbiting Saturn). It could still be the case that goats orbit the sun, or that I have an orange box without pears. For instance, if, for whatever reason, an alien spacecraft decided to launch goats out of their spacecraft around Saturn - but having an orange box that doesn't contain pears in itself does not support

goats orbiting saturn. You could add something to connect the dots but in itself it is not sufficient.

Using the definition of hasty generalization fallacies and logical validity, we corrected each of the statements in our stimulus set to make sure that they were either logically invalid hasty generalization fallacies or logically valid ending up with four logically invalid statements and four logically valid statements for each topic (40 total) (see Table 2 for examples). In order to eliminate linguistic markers that might give the probability of logical validity away (e.g. "Studies show.." is more positively correlated with logical validity than "French gamer Julien Barreaux located..."), we eliminated names and words like "researchers show", "most studies", and "according to published articles". The resulting statements did not have any significant linguistic differences in terms of Word Count, Flesch-Kincaid Grade Level, and sentiment.

4.2 Explanation Feedback

Since logical validity is determined by whether a statement conclusion follows from its premises, the AI explanation feedback templates for causal AI explanations and AI-framed Questioning explanations were shaped in a way that it identifies and highlights the link between premises and conclusion (for examples see table 2). The explanation feedback conditions and shape are defined as follows:

(1) **Causal AI-Explanation**: The AI system gives a reason for why the label is logically valid or logically invalid. "If *reason* then it follows that *label*" for the logically valid statement and "If *premises* then it does not follow that *claim*" for the logically invalid statement.
(2) **AI-Framed Questioning**: AI system asks participants about the causal link between a reason and the system label. It takes a similar form as the causal AI explanation but does not make it clear whether the label actually follows from the reason. "If *premise* does it follow that *claim*?" for both the logically valid and invalid statements.
(3) **No-Explanation**: The AI system does not provide any explanations or feedback of any forms at all.

To generate the causal AI explanations in our study, we used the large language model "GPT-3"[6]. Here, we first gave it a few examples of arguments with hand-crafted causal AI-explanations following the template structure above. We then had it generate causal AI explanations for each argument and manually checked them for accuracy and consistency. We then did the exact same procedure for the AI-framed Questioning explanations and manually checking that there were no linguistic differences between causal AI-explanations and the AI-framed Questioning explanations other than the argument specific reason and label. While we used GPT-3 for this task, we believed that it could easily be done using a rule-based approach when reason and label is known.

4.3 Participants

Participants were recruited from Prolific, an online research participant pool. The total number of participants that enrolled in our study was 234 people. All participants were from the United States and fluent in English with a balanced sex distribution (50% female and 50% male), a mean age of 35.3 years and being 71.2% white. The final number of participants was 204, after excluding the individuals who failed our attention checks or contain missing ratings on prior beliefs of statement topics. Participants were randomly assigned to each condition with the following distribution across conditions: control = 62, causal AI-explanations = 63, and AI-framed questioning = 79, and could complete the study either on their phone, tablet, or computer.

4.4 Procedure

First, participants provided their consent and demographic information (see Figure 3) once enrolled in the study.

Second, participants rated on their prior beliefs and prior knowledge for each topic of the statements used in the study (see section 4.1) from 1-7 (1 = not at all, 7 = very much). They were then randomly assigned to one of the three conditions: (1) No-Explanation, (2) Causal AI-Explanation, and (3) AI-framed Questioning.

Third, to ensure that the participants understood the concept of "logical validity", prior to performing the statement evaluation task, participants were given a one page description of logical validity in layman's terms with examples.

Fourth, the participant entered the main task where they were presented with 10 statements sampled from the 40 total statement dataset of logically valid and logically invalid statements in an random order. This statement evaluation task was based on prior research on a wearable AI system that supports the human reasoning process [11]. For each statement, participants were presented with feedback based on their assigned study condition:(1) a causal AI explanation, or (2) an AI-framed Questioning, or (3) no explanation or any feedback at all. To ensure that the participants read the entire statement, the participants needed to click "next " after reading each statement for the AI feedback to appear with a "slide up" animation. After reading feedback, participants were then asked to discern the logical validity of each statement that they were presented with, to report their confidence in their discernment rating of validity, and to rate whether sufficient information was given in the statement to say that [the claim] is true (1 = not at all, 7 = very much). Below are the questions used in the survey for each statement:

(1) Do you think the statement is logically valid or invalid? (Yes/No)
(2) How confident are you in your rating of logical validity? (on a scale of 1-7: 1 = Not at all, 7 = Very Much).
(3) Is sufficient information given in the statement to support [the claim of the statement]? (on a scale of 1-7: 1 = Not at all, 7 = Very Much)

After the discernment task, participants would be asked to fill out the post-task questionnaires on "cognitive reflection test (CRT)" and "trust in AI" questionnaire.

4.5 Measurements

4.5.1 Weighted Discernment of Logical Validity. For each statement, we calculate a weighted discernment score that aggregates the raw 2-point discernment accuracy ("Correct"/"Incorrect") with the accompanying confidence level (a scale of 1-7: 1 = Not at all, 7 = Very Much). The confidence will be weighted in such a way that a confidence rating like "1", will bring the weight the rating of

Statement	Logical Validity	Causal AI-Explanation	AI-framed Questioning
Violent video games causes people to be aggressive in the real world. A gamer stabbed another after being beaten in the online game Counter-Strike.	Invalid	If one person played video games and was aggressive, ***it does not follow*** that everyone that plays violent video games will be aggressive.	If one person played video games and was aggressive ***does it follow*** that everyone that plays violent video games will be aggressive?
In the United States, racial stratification still occurs. The racial wealth gap between African Americans and White Americans for the same job is found to be a factor of twenty.	Valid	If the racial wealth gap between African Americans and White Americans for the same job is a factor of twenty in the United States, ***it follows*** that racial stratification still occurs.	If the racial wealth gap between African Americans and whites for the same job in the United States is found to be a factor of twenty, ***does it follow*** that racial stratification still occurs?

Figure 2: Example AI Explanations

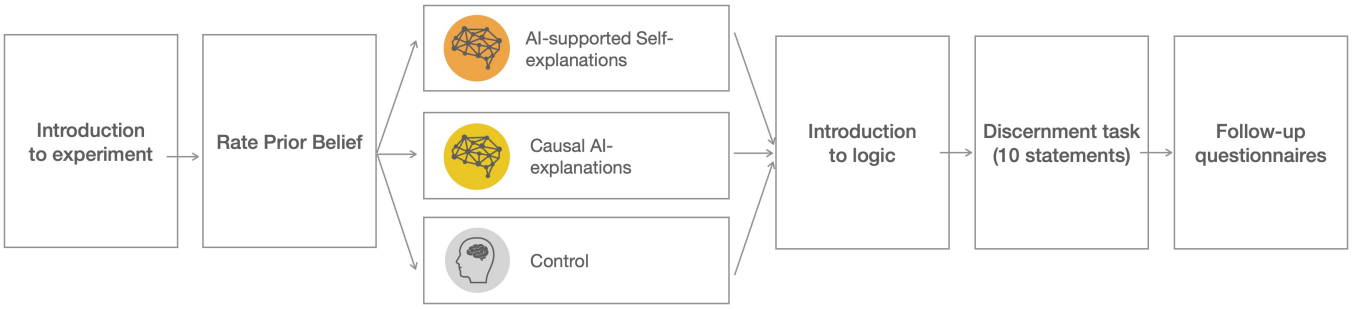


Figure 3: Overview over the experimental procedure.

logical validity to 0.5 (the neutral middle), while a confidence of "7" will keep the rating at either invalid (0) or valid (1). We used the following formuli to calculate weighted discernment accuracy. First we calculate the discernment accuracy:

$$DiscernmentAccuracy = 1 - ||Validity_{rating} - Groundtruth||$$

Next, calculate the weighted factor of confidence from the confidence rating (0.5-"No confidence" to 1-"Fully confident").

$$WeightedFactor = 0.5 * (1 - \frac{Confidence_{rating} - 1}{6})$$

Finally subtract the weighted factor of confidence from the discernment accuracy:

$$WeightedAccuracy = DiscernmentAccuracy - WeightedFactor$$

The weighted discernment score becomes a continuous variable and has a range of 0-100.

4.5.2 Perceived Information Insufficiency. We first measure the perceived information sufficiency through the self-reported scoring from 1-7 (1 = not at all, 7 = very much) on the question "Is sufficient information given in the statement to support [the claim of the statement]?". In analysis, we invert 1-7 scale to report on "perceived information insufficiency" for more a convenient interpretation: a score of 1 indicates that participants find sufficient information is given to support the claim and thus are satisfied with the given information, while a score of 7 indicates that participants find information is insufficient to support the claim and thus are more likely to seek further information to validate the claim.

4.5.3 Cognitive Reflection. To measure the level of critical thinking of subjects, we used cognitive reflection test (CRT), a task designed to measure a person's ability to reflect on a question and resist reporting the first response that comes to mind [13]. For the CRT we randomly sampled three items from the extended CRT [55].

4.5.4 Trust in AI. Finally, following Epstein et al. [12], participants answered a battery of six trust questions derived from Mayer, Davis, and Schoorman [40]'s three factors of trustworthiness: Ability, Benevolence and Integrity (ABI). Previous work, has found that

the six ABI questions are highly correlated with trust ($\alpha = 0.821$), allowing for a single measure of trust that explains 65.3% of the overall variance [12].

4.5.5 Prior Belief and Knowledge. We measured the subject's prior belief about a topic through a self-report scoring from 1-7 (1 = not at all, 7 = very much) on the question "Do you believe that [topic]?". For example, "Do you believe that [violent video games cause aggression]?" Similarly, prior knowledge is measured by 1-7 (1 = not at all, 7 = very much) on the question "Do you have knowledge that [topic]?"

4.6 Approvals

This research has been reviewed and approved by the MIT Committee on the Use of Humans as Experimental Subjects, protocol number E-4115. The research questions and methodology has been pre-registered as "Human-AI Self-explainability" with protocol number #94860 via https://aspredicted.org/.

4.7 Analysis

The purpose of this experimental study is to examine the effects of causal AI-explanations and AI-framed Questioning in supporting human discernment of logical validity. For "Logically Valid" or "Logically Invalid" statements (based on the pre-defined logical validity of the statement stimuli by design), a multivariate Analysis of covariance (MANCOVA) was conducted to examine the main effects of intervention conditions (Causal AI-explanation, AI-framed Questioning, No-explanation) on participants' weighted discernment accuracy (range: 0-100), perceived information insufficiency (range: 1-7), while controlling personal factors (i.e. prior belief and prior knowledge for any statement topic, trust in AI, cognitive reflection) as covariates. Further post hoc tests with Benjamini-Hochberg correction were conducted to identify how intervention conditions differ from each other[39].

5 RESULTS

Findings of valid and invalid statements are reported separately in the following sections.

For invalid statements, MANCOVA results revealed an overall significant main effect of the intervention conditions on the weighted discernment accuracy ($F(2, 1007) = 15.3$, $p < .001 < .05$) and the perceived information insufficiency ($F(2, 1007) = 5.0$, $p = .007 < .05$) after controlling for the effects of personal factors (i.e. prior belief and knowledge, trust in AI, cognitive reflection). Furthermore, several covariates were found to be significant predictors of our two dependent variables, meaning they significantly adjusted the relationship between interventions and the two dependent variables. For example, the weighted discernment accuracy was significantly affected by prior belief ($F(1, 1007) = 6.9$, $p = .009 < .05$) and cognitive reflection ($F(1, 1007) = 7.9$, $p = .005 < .05$), and the perceived information insufficiency was significantly affected by prior belief ($F(1, 1007) = 22.7$, $p < .001 < .05$), cognitive reflection ($F(1, 1007) = 21.2$, $p < .001 < .05$) and trust in AI ($F(1, 1007) = 18.1$, $p < .001 < .05$). However, prior knowledge as a covariate was not found significant.

For valid statements, MANCOVA results revealed an overall significant main effect of the intervention conditions on the weighted discernment accuracy ($F(2, 1007) = 8.4$, $p < .001 < .05$) and the perceived information insufficiency ($F(2, 1007) = 11.3$, $p < .001 < .05$) after controlling for the effects of various personal factors. Additionally, prior belief significantly affected the weighted discernment accuracy ($F(1, 1007) = 7.4$, $p = .007 < .05$) and the perceived information insufficiency ($F(1, 1007) = 17.2$, $p < .001 < .05$).

In summary, these findings indicate that the types of interventions have a significant main effect on the weighted discernment accuracy and the perceived information insufficiency across valid and invalid statements after controlling various personal factors.

5.1 Humans cannot identify logical fallacies very well on their own

We investigated the degree to which participants were able to discern the logical validity of statements and found that without assistance of any AI feedback. The participants' raw discernment accuracy (Mean = 44% accuracy, SD = 26) were lower than the random guess success rate between valid or invalid (50% accuracy) when they were evaluating invalid statements, meaning that their responses were close to simply guessing, while participants supported by causal AI explanations or AI framed questioning achieved a raw discernment accuracy of 57%, 67% respectively. Detailed MANCOVA findings below will present the significant differences between the three intervention conditions.

5.2 AI framed questioning helps improve discernment best

When evaluating invalid statements, after controlling for covariates, both the AI framed questioning condition (mean = 62.5, Std. Error = 1.9) and the causal AI explanation condition (mean = 55.0, Std. Error = 2.2) have a significantly better weighted discernment than the control condition (mean = 46.9, Std. Error = 2.1), with $p(questioning - control) < .001 < .017$ and $p(causal - control) = .007 < .025$ respectively. Moreover, those supported by AI framed questioning also discerned significantly better than those supported by causal AI explanations $p(questioning - causal) = .009 < .05$. Note that the original 0.05 critical value of significance has been adjusted using Benjamini-Hochberg correction to 0.017 for the first rank comparison, .025 for the second rank comparison, and to 0.05 for the third rank comparison among the 3 pairwise posthoc group comparisons.

When evaluating valid statements, after controlling for covariates, both the AI framed questioning condition (mean = 74.7, Std. Error = 1.6) and the causal AI explanation condition (mean = 78.2, Std. Error = 1.7) have a significantly better weighted discernment than the control condition (mean = 68.2, Std. Error = 1.8), with $p(questioning - control) = .007 < .025$ and $p(causal - control) < .001 < .017$ respectively. However, the two AI intervention conditions did not differ significantly from each other in the weighted discernment accuracy, $p(questioning - casual) = .126 > .05$. (adjusted by Benjamini-Hochberg correction).

In general, both AI framed questioning and causal AI explanations helped participants discern significantly better than no feedback. In particular, when encountering fallacies, participants discerned better with AI framed questioning than those with causal



Figure 4: The interface for displaying feedback to participants. Left: No-Explanation, Center: Causal AI-explanations, Right: AI-framed Questioning

AI explanations. In other words, AI framed questioning helps individuals discern best regardless of personal factors.

5.3 Getting causal AI explanation feedback lowers the perceived information insufficiency

When evaluating invalid statements, after controlling for covariates, only the causal AI explanation condition (mean = 4.4, Std. Error = 0.1) has a significantly lower perceived information insufficiency than the control condition (mean = 4.9, Std. Error = 0.1), $p(causal - control) = .002 < .017$

When evaluating valid statements, after controlling for covariates, the causal AI explanation condition (mean = 4.4, Std. Error = 0.1) has a significantly lower perceived information insufficiency than the control condition (mean = 4.9, Std. Error = 0.1), $p(causal - control) < .001 < .017$, and the AI framed questioning condition, $p(causal - self) < .001 < .025$ (adjusted by Benjamini-Hochberg correction).

Such a finding suggests that individuals tend to find the given information is sufficient enough to support the claim (as measured by a significantly lower perceived information insufficiency) when their judgement is corroborated by a second opinion from AI in the causal explanation form. In other words, when supported by causal AI explanations, individuals are more likely to be satisfied with the given information and potentially would not seek further information to verify the claim.

5.4 Personal factors play roles in the weighted discernment accuracy and the perceived information sufficiency

MANCOVA also revealed that several personal factors were found to be significant predictors of participants' weighted discernment accuracy and perceived information sufficiency.

For invalid statements, a weaker prior belief ($F(1, 1007) = 6.9$, $p = .009 < .05$) or a higher cognitive reflection ($F(1, 1007) = 7.9$, $p = .005 < .05$) is significantly associated with a higher weighted discernment accuracy. Additionally, a weaker prior belief ($F(1, 1007) = 22.7$, $p < .001 < .05$) or a higher cognitive reflection ($F(1, 1007) = 21.2$, $p < .001 < .05$) or a lower trust in AI ($F(1, 1007) = 18.1$, $p < .001 < .05$) is significantly associated with a higher perceived information insufficiency.

For valid statements, a greater prior belief is significantly associated with a higher weighted discernment accuracy ($F(1, 1007) = 7.4$, $p = .007 < .05$) and a lower perceived information insufficiency ($F(1, 1007) = 17.2$, $p < .001 < .05$).

Overall, these significant effects about personal factors as covariates from MANCOVA suggest the two AI interventions have a main training effect in improving discernment accuracy despite personal

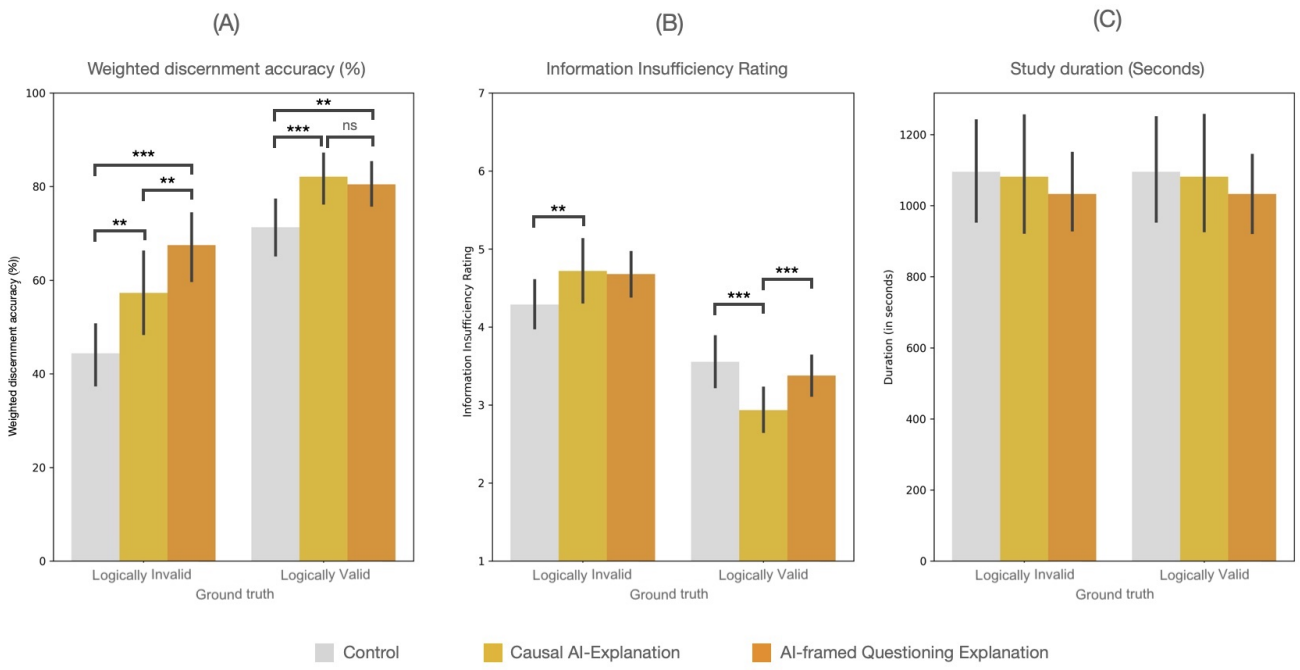


Figure 5: Overview of the effects of AI systems on human discernment of logically valid and invalid statements. (A) Weighted discernment accuracy for the different feedback types on logically valid and invalid statements. (B) The inverted users' rating of information being insufficient to rate the claim as true for the different feedback types on logically valid and invalid statements. (C) Time for users to complete the study for the different feedback types. $^{*}<0.05$, $^{**}<0.01$, $^{***}<0.001$, adjusted using the Benjamini-Hochberg correction.

factors. In other words, individuals can benefit from AI interventions regardless of their prior belief, high or low cognitive reflection levels, or high or low trust in AI. Additionally, there are a few implications regarding how these personal factors affect discernment and perceived information insufficiency when encountering fallacies. Firstly, a strong prior belief affect discernment differently for valid versus invalid statements differently, undermining the judgment for invalid statements while strengthening the judgment for the valid ones. Secondly, individuals with traits such as a weaker prior belief, a higher cognitive reflection (indicating critical thinking that override the first response that comes to mind) and a lower trust in AI are more likely to discern better and find the given information insufficient, and potentially seek alternative sources. In contrast, individuals with a strong prior belief, a lower cognitive reflection (driven by intuition), a higher trust in AI are more likely to be satisfied with information given at hand thus be posed to risks in discern correctly. For example, these risks include misleading information presented or trusting AI feedback without double-checking other sources of evidence. Further research using a mixed model analysis is needed to understand the detailed interaction effects of these personal factors together with interventions.

5.5 Qualitative Analysis

Here we report the participants' subjective experience during the discernment task in the three conditions. Each quote below is from different participants in the condition. To sample the thinking processes of the participants, we asked them to report their internal thinking process. These results complement participants' ratings with narratives. An inductive coding [5] was conducted to discover salient themes. Two coders iteratively coded the narratives with themes, discussed and aligned any disagreements, and added emerging themes until reaching theoretical saturation [10].

5.5.1 Participant's Experiences in the Control group. As reflected in the performance result of the discernment task, participants in the control without the intervention found it difficult to discern logically valid statements from the logically invalid statements. Without the AI feedback, most participants reported that they mostly relied on their feelings, intuitions, or prior knowledge rather than using reflective thinking in determining the validity of the statement:

> "I didn't have a specific thinking process. I just went with the knowledge I already know and what each statement was saying."

> "There was not that much of a thinking process, all 10 statements made perfect sense"

Though, there are also few participants that in their writing show their critical thinking process when evaluating the statement, some try to evaluate the logical consistency within the statement, while some look for evidence and fact to support the claim:

> "I tried to figure out if the second statement [premise] clearly described a truth of the first statement [claim]. If so that would be logically valid. If the second statement didn't clearly back the first statement i figured that to be logically invalid." "I was thinking about the two sentences and how the second sentence supports the first (if at all)."
>
> "it was okay i guess. I just said the statement out loud to myself and looked for proof in the statement"

5.5.2 Participant's Experiences in the Causal AI Explanation group. For the causal AI explanation condition, participants generally reported the usefulness of the AI feedback. However, participants with low critical reflection were found to have a tendency to be persuaded by the AI explanations and go with the AI feedback:

> "My thinking process was typically to go with the feedback given. I like that the feedback provided an explanation as to why it made its ruling"
>
> "It was interesting to get a second opinion and that persuaded my position to a certain extent."

The fact that participants are willing to give up their thinking and follow the AI recommendation is a concerning issue that some participants also noticed by themselves, as reported here: "I thought it was useful, made me think more carefully. But I also worried [me] that I was giving to much weight to the feedback." Participants reported that they felt more confused when they disagreed with the AI explanations and more confirmed when they agreed with the AI explanations. This further demonstrates the notion of AI explanations potentially overpowering the human thinking:

> "I feel confused and doubt myself when I disagree with the AI"
>
> "The feedback helped confirm my own thoughts on the validity of the statements"

Finally, one emerging theme of the responses from participants with low cognitive reflection is that, even though the feedback on the AI on the logical connection is sufficient for the task, they felt the feedback to be limited, and that the needed more information:

> "I liked the feedback but I wish there had been more evidence as well."
>
> "The feedback was limited. My thinking process was to fall back on what I already know and to do further research on the topic if I desire to do so."

On the other hand, participants with high level of cognitive reflection were doubting the accuracy of AI feedback, and tended to disagree when they thought the AI system was giving them biased or incorrect feedback — although the feedback was 100% accurate:

> "It was interesting to see the AI made assumptions based on the information given. However, I know that AI's often times think in a vaccuum, and [they] are not affected by nuance and real world scenarios. When considering the issues, I often thought about what other world factors affected the statement, and even if something seemed "logically sound" (I.E. the statement in itself made sense) whether or not the information given could be backed up. Anyone can say "statistics show" and make what sounds like a logical statement, but those statistics need to be analyzed and discussed to rule out any bias or skewing."

Finally, in both groups of high and low cognitive reflection, some participants reported their hesitation and resistance to follow the feedback, echoing the feedback of the control condition, where subjects would just go with their intuition:

> " It was interesting to see what the AI logic said, but I tried to go with what I felt without the influence of AI"
>
> "It was helpful to get feedback to help me make a better decision. But in the end, my decision was based on how logical the statement felt to me, even if the AI disagreed."

5.5.3 Participant's Experiences in the AI-framed Questioning group. We found that participants in the AI-framed Questioning condition were generally having positive experiences and found the feedback in terms of questioning to be helpful regardless of their cognitive reflection level. Participants with low cognitive reflection in particular, reported how the question of the AI system helped re-frame the statement, and hence making it easier for them to understand the logical connection or disconnection between the claim and the premise:

> The AI feedback was helpful, and it framed the presented information in a way that was logical and easy to digest. I evaluated whether a given statement seemed logical given the information provided.
>
> "The feedback provided by the AI gave a clearer picture as to what the statement was proclaiming. I was able to process the question asked by the AI if I had trouble determining how a statement listed was logically valid or not."

Participants further reported that after receiving the feedback, the questions helped them reflect and update their decisions.For the participants with high level of cognitive reflection, participants reported that they already applied their reasoning process on the information, and the question from the AI made them reflect on their reasoning process:

> "It was nice to have an AI guide me by interpreting the information presented and ask reasonable questions based on its interpretation. However, I tried to double-check the original statements a few times to make sure the AI feedback was valid. It made decision-making easier overall and lent me some confidence in my answers."
>
> "The feedback helped me to understand how the two statements were connected and what they implied. My thinking process mainly focused on finding another possible factor that could be left out of the statements. I also thought that circumstances that showed one example of something, like with the circumcision case, were invalid while trends were valid."

In both groups, after receiving recurring questions from the AI system, users also reported that they learned to recognize the patterns of hasty generalizing fallacies (the invalid statements), highlighting how they integrated and internalized the AI feedback into their own thinking:

> "The AI feedback helped a lot, because I realized that many of the statements only had one example to support a claim, but the AI brought up that one instance does not mean all. "
>
> "The feedback questions helped a lot by simply rephrasing and asking whether a statement's premise and conclusion made sense. My thought process was mainly to avoid generalizing from a single event, and only generalizing from trends and research statistics."

6 DISCUSSION

The study showed that both AI framed questioning and causal AI feedback on the logical structure of information can significantly improve human discernment accuracy of logical validity over a human alone, regardless of any personal factors such as prior belief, cognitive reflection or trust in AI. This is promising as it shows that people with strong or weak prior belief, high or low cognitve reflection, high or low trust in AI can all benefit from AI interventions in improving their discernment.

Moreover, the AI framed questioning led to the a better discernment than causal AI feedback when encountering fallacies. This aligns with prior literature on AI-assisted reasoning that shows the potential of technology to support critical thinking [11]. This is also consistent with previous finding that showed how cognitive forcing functions can increase a subject's tendency to reflect on AI feedback and use it more effectively (with higher accuracy) [7]. Beyond cognitive forcing functions that omit AI feedback labels, our results show that questions can go even further in terms of effectiveness. Based on our qualitative data, we hypothesize that this is because of questions being empty (not having any truth value) which makes the user need to decide for themselves.

Additionally, participants with causal AI explanations were significantly more satisfied with the given information in support of the claim (indicated by lower perceived information insufficiency) compared to those who are not supported with any feedback at all or supported with AI framed questioning. This could expose them to risks of their erroneous prior beliefs or their lack of critical thinking without double-checking extra sources.

6.1 Generalizability of framed-questioning explanations

Our results demonstrate the effects of AI-framed Questioning on human logical discernment of socially divisive statements. However, beyond logical discernment, what kinds of decision-making tasks can use the AI-framed Questioning approach? While it is an empirical question whether the method yields similar performance gains in other domains, there are certain inherent limitations of the approach. For instance, the AI-framed Questioning approach assumes that the AI model is able to causally link a reason to a recommendation, and as such any AI system capable of generating causal AI explanations that logically connect to a prediction label could use this method. For example, the framed-questioning method could be used to explain the prediction of an AI system that is making a medical diagnosis such as "If the patient has X symptoms, does that mean that necessarily mean that they have Y illness?", assisting the user with connecting the symptoms to the diagnosis. However, not all AI models are able to derive causal links to their predictions, such as some attention-based deep neural networks, and in such cases the method might not work adequately. In the future, we plan to evaluate the framed-questioning method in other domains and tasks to further explore its generalizability.

Moreover, our experiment compared AI-framed Questioning to always correct Causal AI-explanations. One might argue that this is problematic because previous work on over-reliance on AI systems usually identifies the issue to be that explanation increases people's reliance when the AI system is wrong, and since our experiment does not compare AI-framed questioning with an AI system that can be wrong, the results do not contribute much to discussions on over-reliance. However, we argue questioning might help prevent over reliance considering that it does not feed and tell users direct answers but provoke them to reflect by themselves. Since AI-framed Questioning is prediction agnostic and always gives the same question feedback — whether or not a statement is logically valid or not — the user is unable to rely on on the prediction of the AI system (there simply are no answers given). Further study and measures are necessary to examine over reliance of questioning AI and compare properly with always and not always correct AI

6.2 The future of AI interfaces that ask instead of tell

This paper presents the idea of AI-framed Questioning as a valid AI-feedback modality inspired by the ancient method of Socratic questioning that uses intelligently formed questions to provoke the human reasoning process, allowing the user to correctly discern logical validity of statements by themselves. This method emphasizes the role of AI systems in supporting human critical thinking by providing a scaffold for the thinking process and making it less effortful for humans to draw logical consistency within statements. Our experiment demonstrates an opportunity for a new style of human-AI interaction that encourages the person to evaluate and make decision for themselves rather than taking feedback from AI systems at face value. We foresee various future applications for AI-framed Questioning, such as enhanced critical thinking about social media recommendationsor with doctors and policy makers where critical thinking is paramount.

6.3 Limitations and Future Work

Though the idea of AI-framed Questioning is promising, our work opens up many doors for improvements and future research.

6.3.1 Balancing between telling and asking. In this paper we highlight the power of "asking the right question" to trigger or engage reflective thinking. However, AI-framed Questioning might not always be appropriate. First, correct discernment when getting AI-framed Questioning relies somewhat on the user's basic ability to connect the dots and see logical contradictions or absurdities in the explanations. However, it is not always guaranteed that the questions in the AI-framed Questioning can be correctly understood

or interpreted by the user. Secondly, the user might not always want to be questioned by the AI system. There might be some cases where merely being told the answer with causal AI-explanations is sufficient — either because the user can't figure out the answer for themselves or they do not have the time or desire to think reflectively. Perhaps they have over time learned the weaknesses of the AI system and feel completely fine relying on the causal AI-explanation. Humans need to know when to rely and when not to rely. They can learn this over time[2, 65]. Thus, understanding the context, balancing and understanding when to ask and when to tell is a challenge to make an effective human-AI system work in the real world.

6.3.2 Administrating Cognitive Resources. A limitation that has not been measured in the experiment is the differences in cognitive load imposed by the different AI explanations. Since AI-framed Questioning are posed as questions which potentially require more reflection than being told something, cognitive resources required from the individual to process the the AI-framed Questioning might be higher. Rather than optimizing only for accuracy, it is also important to consider that AI-support systems that are effectively easier to use may in some context be desirable.

6.3.3 Exploring multiple types of questions. The questions used as the intervention in our study were generated by a rule-based approach ("If [reason] does it follow that [label]?"). However, there are other kinds of questions that can be used to promote critical thinking. For instance, Socrates used multiple self-reflection questions in multiple turn conversation. This could be an inspiration for future research that explores how AI-framed Questioning could generate multiple turn dialogues between a human and AI systems. Such approaches have already been tested with systems that teach critical thinking [31]. Merging this work with AI systems could yield interesting and potentially impactful results.

6.3.4 Generalizing to other fallacies. Our current study chose to evaluate AI-feedback on hasty generalization fallacies, one of the most common type of logically invalid statements. Participant reports showed that some participants were able to learn characteristics of these fallacies. Future studies should expand to other type of logically invalid statements to make sure that effects are replicable even with more difficult to discern logically invalid statements.

6.3.5 Understanding social factors. As mentioned in the introduction, reasoning is often a social process arising from conflicting ideas in conversation among humans. Using AI as a thought partner requires further investigation into other social factors in reasoning, such as personification of the AI system and how admiration or emotions towards the personification will impact the user's response.

6.3.6 In-the-wild study. Our study was conducted with an online participant pool in a controlled environment. The next step for this study would be to evaluate the effect of AI-framed Questioning in the wild by implementing the system as part of a social media or smart home device to test whether it could positively impact users in the world. We imagine AI systems like these could not just help people deal with filtering out low quality information but also help them improve their own argumentation.

6.4 Conclusion

Despite the importance of critical thinking, research reveals that people's reasoning ability suffers from personal biases and cognitive resource limitations, leading to non-optimal outcomes. This paper presents the idea of AI-framed Questioning, that asks questions to provoke the user to discern the logical validity of information by themselves in contrast to typical AI explanations that rely on the user passively receiving feedback from the AI system. Our study results show that AI-framed Questioning increases discernment accuracy of flawed statements and users' ratings of flawed statements level of sufficient information. While the experiment shows the technique is applicable for the specific task of logical validity, it exemplifies a future type of system where AI agents work together with and challenges humans, instead of simply telling them what they should believe or do. In a world where we are constantly bombarded with information, it is more important than ever to be able to think critically and improve our discernment capabilities beyond our current state.

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